# Peter W. Fields

Physicist | Machine Learning Researcher

Chicago, IL • (646) 599-5151

pfields97@gmail.com • peter-fields.github.io • github.com/peter-fields • Google Scholar


## SUMMARY

- PhD researcher in statistical physics / machine learning with 5+ years of experience building and evaluating probabilistic models (energy-based / graphical / generative models) grounded in theory and validated empirically on large-scale biological sequence data with GPU-accelerated computation.
- Developed forward-pass diagnostic tools for mechanistic interpretability of transformer attention heads; demonstrated statistically significant distinguishability between circuit and non-circuit heads in GPT-2's IOI circuit ($p<0.001$), without requiring activation patching.
- End-to-end research experience—problem formulation, statistical modeling, and communicating results—with cross-disciplinary collaborations across biology, neuroscience, and AI.

## EDUCATION

**PhD in Physics, University of Chicago** – March 2026
Advisor: Stephanie E. Palmer

**Bachelor of Science in Physics, The City College of New York** – 2019

## PUBLICATIONS & PREPRINTS

**Understanding temperature tuning in energy-based models**
**P. W. Fields**, V. Ngampruetikorn, D. J. Schwab, S. E. Palmer.
Preprint (2025). arXiv:2512.09152. *under review at Physical Review Research.*
Code: github.com/peter-fields/temp-tune

**Understanding Energy-Based Modeling of Proteins via an Empirically Motivated Minimal Ground Truth Model**
**P. W. Fields**, V. Ngampruetikorn, R. Ranganathan, D. J. Schwab, S. E. Palmer.
Synergy of Scientific and Machine Learning Modeling Workshop at International Conference in Machine Learning (2023). openreview.net/forum?id=vxn5QGPFyi
*Presented at ICML 2023 workshop.*
Code: github.com/peter-fields/toysector

**Tunable Pseudocapacitive Intercalation of Chloroaluminate Anions into Graphite Electrodes for Rechargeable Aluminum Batteries**
J. H. Xu, T. Schoetz, J. R. McManus, V. R. Subramanian, **P. W. Fields**, R. J. Messinger.
*Journal of the Electrochemical Society* 168, 060514 (2021)

## SELECTED WRITING

**Attention Diagnostics: Testing KL and Susceptibility on the IOI Circuit**
peter-fields.github.io/attention-diagnostics/

**Why Softmax? A Hypothesis Testing Perspective on Attention Weights**
peter-fields.github.io/why-softmax/

## EXPERIENCE

**Palmer Theory Group, University of Chicago** (Jan 2021 – Present)

- Applied information-theoretic and physics-informed methods across projects in protein modeling and biologically motivated neuronal-like systems; developed working fluency in computational neuroscience and quantitative biology.
- Developed synthetic benchmarks and exact maximum likelihood methods to study finite-sample effects and generalization in protein sequence energy-based models.
- Characterized KL divergence structure to understand generative model failure modes.
- Built attention diagnostic pipeline in TransformerLens (Python); tested KL divergence and susceptibility metrics across IOI circuit heads in GPT-2.
- Ran large-scale compute jobs on HPC clusters using Slurm, including GPU-accelerated workloads via CUDA.jl in Julia.

**Messinger Research Group, The City College of New York** (Jan 2018 – Aug 2019)

- Physics-based characterization of novel battery technologies
- Designed and executed impedance spectroscopy experiments
- Studied diffusive, kinetic, and electrical properties of battery systems

**Teaching Assistant, General Physics I & II, University of Chicago** (2019–2021)

## SKILLS

**Coding:** Julia (including GPU/CUDA programming via CUDA.jl), Python (PyTorch, TransformerLens), Mathematica, Unix/CLI, LaTeX, Git, Jupyter, Slurm/HPC clusters

**Methods:**

- Information theory; statistical inference; regularization; hyperparameter search; model selection
- Supervised, unsupervised, and generative/probabilistic models
- Graphical models; network/graph structure inference
- Graduate coursework in: Quantum Field Theory, General Relativity, Advanced Statistical Mechanics, Biophysics, Computational Neuroscience

## SELECTED TALKS & PRESENTATIONS

***Temperature-tuning trained energy functions improves generative performance***
APS Global Physics Summit 2025 – **P. W. Fields**, V. Ngampruetikorn, D. J. Schwab, S. E. Palmer

***Characterizing Inference of Non-reciprocal Connections in the Kinetic Ising Model***
APS March Meeting 2024 – **P. W. Fields**, Cheyne Weis, S. E. Palmer, Peter Littlewood

## AWARDS & AFFILIATIONS

**Soft and Living Matter Exchange Award,** Institute for Complex Adaptive Matter (Dec 2023)
Funded 3-week exchange to work with a collaborator, David J. Schwab
**Center for Physics of Evolving Systems Fellowship,** University of Chicago (Jan 2022–Dec 2023)
For investigators at the interface of physics and biology, focusing on principles of the evolution of biological function.

**Affiliated with**: NSF-Simons National Institute for Theory and Mathematics in Biology, Center for Living Systems at the University of Chicago, CUNY–Princeton Center for the Physics of Biological Function
CUNY–Princeton Center for the Physics of Biological Function